

ASCHAM SCHOOL

2000 TRIAL HSC EXAMINATION

MATHEMATICS 3 / 4 UNIT COMMON PAPER

Time allowed: 2 hours

- All questions should be attempted.
- All necessary working must be shown.
- All questions are of equal value.
- Marks may not be awarded for careless or badly arranged work.
- Write your name and the number of the question on each booklet.
- Begin each question in a new booklet.
- Approved calculators may be used.

Stephanie Sim 2

QUESTION 1

- a) Differentiate i) $4\sec^3 x$
ii) $\sin^{-1} \frac{1}{2}x$ (2)

- b) Find the primitive of $\frac{1}{9x^2+1}$ (2)

- c) Find the co-ordinates of the point P which divides the line joining A(-3,4) and B(2,-8) externally in the ratio 2:5. (2)

- d) Use the substitution $x = \cos \theta$ to evaluate

$$\int_{\frac{1}{2}}^1 \frac{\sqrt{1-x^2}}{x^2} dx \quad (3)$$

- e) If α , β and γ are the roots of the equation $x^3 - 2x + 5 = 0$, find the value of $\alpha^2 + \beta^2 + \gamma^2$ (3)

QUESTION 2

- a) $x = 0.8$ is a good approximation to a root of the equation $x^2 = \cos x$.

Use Newton's method once to find a better approximation to the root, giving your answer to 2 decimal places. (2)

- b) i) Show that $x = 2$ is a zero of $x^3 - 4x^2 + 8$
 ii) Hence find all the real zeros of $x^3 - 4x^2 + 8$, leaving your answers in simplified surd form.
 iii) Solve for x : $\frac{4}{x-2} \leq x$ (6)
- c) Solve the equation (to the nearest degree)
 $3 \cos x - 4 \sin x = 3$ for $0^\circ \leq x \leq 360^\circ$. (4)

QUESTION 3

- a) Sketch the graph of $y = 3 \cos^{-1}(2x + 1)$. (2)
- b) Solve for x , $0 \leq x \leq 2\pi$, $2 \sin^2 x < \sin x$. (3)
- c) The acceleration of a particle moving in a straight line is given by $\frac{d^2x}{dt^2} = \frac{-72}{x^2}$, where x metres is the displacement from the origin after t seconds. Initially the particle is 9 metres to the right of the origin with a velocity of 4 metres per second.
- i) Show that the velocity v of the particle in terms of x is $v = \frac{12}{\sqrt{x}}$. Explain why v is always positive for the given initial conditions.
 ii) Find an expression for t in terms of x .
 iii) How many seconds (to the nearest second) does it take for the particle to reach a point 35m to the right of the origin? (7)

QUESTION 4

- a) i) Show that $\frac{d}{dx}(\cos^3 x \sin x) = 4 \cos^4 x - 3 \cos^2 x$.
 ii) Hence show that $\int_0^{\frac{\pi}{4}} \cos^4 x dx = \frac{1}{4} + \frac{3\pi}{32}$ (6)
- b) Prove by mathematical induction that $\sum_{r=1}^n r 2^{r-1} = 1 + (n-1)2^n$ (5)
- c) Solve for x if $\cos 2x = \frac{1}{2}$ (1)

QUESTION 5

- a) Find, to the nearest degree, the acute angle between the curves $y = x^2 - 1$ and $y = x(x-1)$ (2)
- b) Without calculus, draw the graph of $y = \frac{x}{(x-1)^2}$, showing asymptotes and intercepts on axes. (2)
- c) Joyce and Agnes are playing a game. Joyce has 6 cards numbered 1 to 6 and Agnes has 8 cards numbered 7 to 14. Joyce goes first and draws a card, looks at it and replaces it in her pack. If it is even, she wins. If it is odd, it is Agnes' turn to draw a card, look at it and replace it in her pack. If Agnes draws an odd card, she wins. If Agnes draws an even card, she loses her turn and Joyce draws a card and so on.
- Find the probability that Joyce wins:
- i) in the first draw
 ii) in her second draw
 iii) in her third draw
 iv) in the long run. (4)
- d) A spherical balloon is being blown up so that its surface area is increasing at the constant rate of $10 \text{ cm}^2/\text{second}$. Find the rate at which the volume is increasing when $r = 5 \text{ cm}$. (4)

QUESTION 6

- a) i) Show that the equation of the tangent to the parabola $x^2 = 16y$ at any point $P(8t, 4t^2)$ on it is $y = tx - 4t^2$.
- ii) Show that the equation of the line l through the focus S of the parabola which is perpendicular to the focal chord through P is $(t^2 - 1)y + 2tx = 4(t^2 - 1)$.
- iii) Find the locus of the point of intersection of the line l and the tangent at P . (7)
- b) AB and CD are two towers of equal heights. CD is due north of AB . From a point P on the same horizontal plane as the feet B and D of the towers, and bearing due east of the tower AB , the angles of elevation of A and C , the tops of the towers, are 47° and 31° respectively. If the distance between the towers is 88m , find the height of the towers to the nearest metre. (5)

QUESTION 7

- a) At the Tildesley Tennis Competition, Felicity served a ball from a height of 1.8m above the ground. The ball was hit in a horizontal direction with initial velocity $V = 35\text{m/s}$. Assume that the equations of motion for the ball in flight are $y = -5t^2 + 1.8$ and $x = 35t$ where the acceleration due to gravity is taken at 10m/s^2 .
- i) How long does it take for the ball to hit the ground?
- ii) How far will the ball travel horizontally before bouncing?
- iii) The net is 0.95 metres high and is 14 metres away from where Felicity hit the ball. Will the ball clear the net? Explain. (5)
- b) Geometry question on the next page.

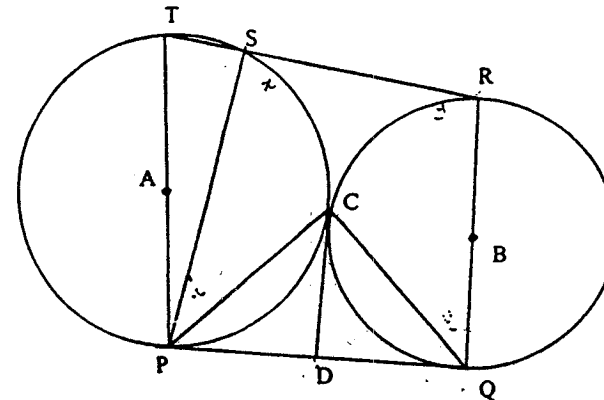
b) **DO NOT RE-DRAW THIS DIAGRAM.**

DETACH THIS PAGE AND STAPLE IT IN YOUR BOOK FOR QUESTION 7.

DO THE WRITING IN YOUR ANSWER BOOK NOT ON THIS PAGE.

A circle centre A touches a smaller circle centre B externally at a point C . PQ is a common tangent to the two circles, touching them at P and Q . CD is a common tangent to both circles at C . RT cuts the circle centre A at S .

- i) Show that $\angle PCQ = 90^\circ$.
- ii) Show that P , C and R are collinear.
- iii) Show that P , Q , R and S are concyclic points. (7)



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84
81
cos²x

Stephanie Sim

Excellent work!

$$1) \quad A.) \quad i.) \quad \frac{d}{dx} 4 \sec^3 x = 12 \sec^2 x \times \frac{1}{\cos^2 x} \times \sin x$$

$$= 12 \sec^2 x \tan x \sec x \quad \checkmark$$

$$= 12 \tan x \sec^3 x.$$

$$ii.) \quad \frac{d}{dx} \sin^{-1} \frac{x}{2} = \frac{1}{\sqrt{1 - \frac{x^2}{4}}} \times \frac{1}{2} = \frac{x}{\sqrt{4 - x^2}} \times \frac{1}{x} = \frac{1}{\sqrt{4 - x^2}} \quad \checkmark$$

$$b) \quad \int \frac{1}{9x^2 + 1} dx = \int \frac{1}{9(x^2 + \frac{1}{9})} dx = \frac{1}{9} \int \frac{1}{x^2 + \frac{1}{9}} dx \quad \checkmark$$

$$= \frac{1}{9} \tan^{-1} 3x + C = \frac{1}{3} \tan^{-1} 3x + C \quad \text{(Must check with standard integrals table)}$$

$$c) \quad P = \left(\frac{2(2) - 5(-3)}{-3}, \frac{2(-8) - 5(4)}{-3} \right) \quad \checkmark$$

$$= \left(\frac{4 + 15}{-3}, \frac{-16 - 20}{-3} \right) = \left(\frac{-19}{3}, \frac{12}{-3} \right) \quad \checkmark$$

$$d) \quad \int_{\frac{1}{2}}^1 \frac{\sqrt{1-x^2}}{x^2} dx$$

$$x = \cos \theta$$

$$\frac{dx}{d\theta} = -\sin \theta$$

$$dx = -\sin \theta d\theta$$

$$= \int_{\frac{\pi}{3}}^0 \frac{\sqrt{1-\cos^2 \theta}}{\cos^2 \theta} \times -\sin \theta d\theta$$

$$= \int_{\frac{\pi}{3}}^0 \frac{-\sin^2 \theta}{\cos^2 \theta} d\theta = \int_{\frac{\pi}{3}}^0 -\tan^2 \theta d\theta \quad \checkmark$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$\tan^2 \theta = \sec^2 \theta - 1$$

$$= \int_{\frac{\pi}{3}}^0 -\sec^2 \theta + 1 d\theta$$

$$= (-\tan \theta + \theta) \Big|_{\frac{\pi}{3}}^0$$

$$= 0 - \left(-\tan \frac{\pi}{3} + \frac{\pi}{3} \right) = +\tan \frac{\pi}{3} - \frac{\pi}{3}$$

$$= \sqrt{3} - \frac{\pi}{3} \quad \checkmark$$

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E.) Let $P(x) = x^3 - 2x + 5 = 0$

$$\alpha + \beta + \gamma = 0 \quad \alpha\beta + \alpha\gamma + \beta\gamma = -2 \quad \checkmark$$

$$\alpha\beta\gamma = -5 \quad \checkmark$$

$$\begin{aligned} \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= 0 - 2(-2) = \underline{4} \end{aligned}$$

2) A.) $x^2 = \cos x$

$$x^2 - \cos x = 0$$

Let $P(x) = x^2 - \cos x = 0$

$$P'(x) = 2x + \sin x \quad \checkmark$$

$$x = 0.8$$

$$x_1 = x - \frac{P(x)}{P'(x)} \quad \checkmark$$

$$x_1 = 0.8 - \frac{P(0.8)}{P'(0.8)} = \underline{0.82 \text{ (2dp)}} \quad \checkmark$$

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B) i.) Let $P(x) = x^3 - 4x^2 + 8$

$$P(2) = 2^3 - 4(2^2) + 8 = 8 - 16 + 8 = 0$$

$\therefore x=2$ is a zero of $P(x)$

ii.) If $x=2$ is a zero of $P(x)$, $\therefore (x-2)$ is a factor of $P(x)$

$$\begin{array}{r} x^2 - 2x - 4 \\ (x-2) \overline{) x^3 - 4x^2 + 0x + 8} \\ \underline{x^3 - 2x^2} \\ -2x^2 + 0x \\ \underline{-2x^2 + 4x} \\ -4x + 8 \\ \underline{-4x + 8} \\ 0 \end{array} \quad \checkmark$$

$$\therefore P(x) = (x-2)(x^2 - 2x - 4)$$

$$x=2 \text{ OR } x^2 - 2x - 4 = 0$$

$$x = \frac{2 \pm \sqrt{4 - 4(-4)}}{2}$$

$$x = \frac{2 \pm \sqrt{20}}{2}$$

$$= 1 \pm \sqrt{5}$$

$\therefore$ the zeroes of $P(x)$ are

$$\underline{x=2, \quad x=1+\sqrt{5} \text{ and } x=1-\sqrt{5}}$$

(2)

-3-

iii) $\frac{4}{x-2} | -x | \leq 1 \Rightarrow \frac{4}{x-2} \leq x$ (multiply throughout by $(x-2)^2$)

$$\frac{4}{x-2} x (x-2)^2 \leq x (x-2)^2$$

$$4(x-2) \leq x(x^2-4x+4)$$

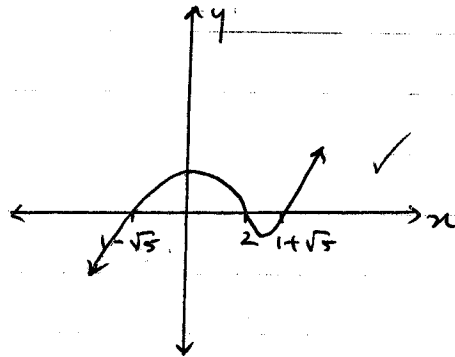
$$4x-8-x^3+4x^2-4x \leq 0$$

$$4x^2-8-x^3 \leq 0$$

$$x^3-4x^2+8 \geq 0 \quad \checkmark$$

This is true for $1-\sqrt{5} \leq x \leq 2$

or $x \geq 1+\sqrt{5}$ $\checkmark$



c) $3\cos x - 4\sin x = 3 \quad 0 \leq x \leq 360^\circ$

$$3\cos x - 4\sin x = R\cos(x+\alpha)$$

$$= R(\cos x \cos \alpha - \sin x \sin \alpha)$$

$$3 = R\cos \alpha \quad \text{--- (1)}$$

$$4 = R\sin \alpha \quad \text{--- (2)}$$

$$\frac{(2)}{(1)} = \tan \alpha = \frac{4}{3} \quad ; \quad \alpha = \tan^{-1} \frac{4}{3} \quad \checkmark$$

$$\alpha = 53^\circ 8'$$

$$R = \sqrt{a^2+b^2} = \sqrt{9+16} = \sqrt{25} = 5 \quad \checkmark$$

$$\therefore 3\cos x - 4\sin x = 5\cos(x+53^\circ 8') = 3$$

$$\cos(x+53^\circ 8') = \frac{3}{5}$$

$$x+53^\circ 8' = \cos^{-1} 0.6 \quad (53^\circ 8' \leq x+53^\circ 8' \leq 413^\circ 8')$$

$$x+53^\circ 8' = 53^\circ 8', 306^\circ 52', 413^\circ 8'$$

$$\therefore x = 0, 253^\circ 44', 360^\circ \quad \checkmark$$

$$\begin{array}{c|c} 5 & 4 \\ \hline 7 & c \end{array} \quad \checkmark$$

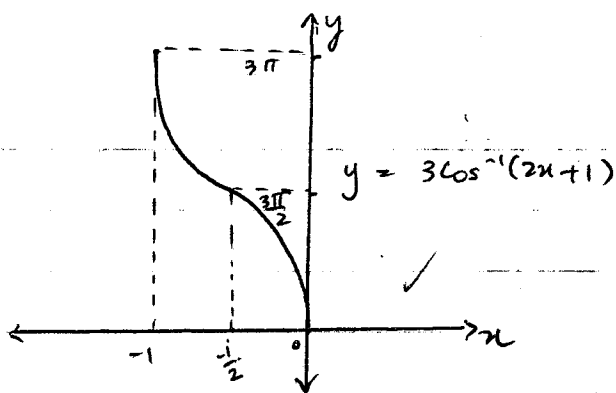
3) A) $y = 3\cos^{-1}(2x+1)$

$D: -1 \leq 2x+1 \leq 1$

$-2 \leq 2x \leq 0$

$-1 \leq x \leq 0$ ✓

Range $0 \leq y \leq 3\pi$



B) $2\sin^2 x < \sin x$

$2\sin^2 x - \sin x < 0$

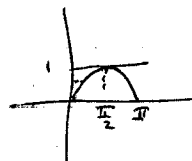
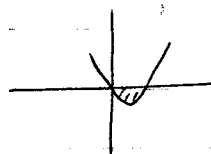
$\sin x (2\sin x - 1) < 0$

$\sin x = 0$ or $\sin x = \frac{1}{2}$ ✓

$\therefore 0 < \sin x < \frac{1}{2}$

$\sin x = 0$ when $x = 0, \pi, 2\pi$

$\sin x = \frac{1}{2}$ when $x = \frac{\pi}{6}, \frac{5\pi}{6}$



$\therefore 0 < x < \frac{\pi}{6}$ or $\frac{5\pi}{6} < x < \pi$ ✓

c) $\ddot{x} = -\frac{12}{x^2}$

i.) $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$; $\frac{1}{2} v^2 = \int -12x^{-2} dx$

$\frac{1}{2} v^2 = \frac{-12x^{-1}}{-1} + C$

$\frac{1}{2} v^2 = \frac{12}{x} + C$ ✓

when $x = 9$, $v = 4$.

$\frac{1}{2} \cdot 16 = \frac{12}{9} + C$; $8 = 8 + C$; $C = 0$

$\therefore \frac{1}{2} v^2 = \frac{12}{x}$ ✓

$v^2 = \frac{144}{x}$; $v = \pm \frac{12}{\sqrt{x}}$. Since $v > 0$ when $x = 9$ and $t = 0$,
 $v = \frac{12}{\sqrt{x}}$ m/s

(3)

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$$ii.) \quad \frac{dx}{dt} = \frac{12}{\sqrt{x}}$$

$$\frac{dt}{dx} = \frac{\sqrt{x}}{12} \quad ; \quad t = \int \frac{\sqrt{x}}{12} dx$$

$$= \frac{2x^{\frac{3}{2}}}{3} \times \frac{1}{12} + C$$

$$t = \frac{x^{\frac{3}{2}}}{18} + C \quad \checkmark$$

When $t=0$, $x=9$.

$$0 = \frac{9^{\frac{3}{2}}}{18} + C$$

$$0 = \frac{27}{18} + C = \frac{3}{2} + C; \quad C = -\frac{3}{2}$$

$$\therefore t = \frac{x^{\frac{3}{2}}}{18} - \frac{3}{2} \quad \checkmark$$

iii.) Find t when $x=35$.

$$t = \frac{35^{\frac{3}{2}}}{18} - \frac{3}{2} \quad \checkmark$$

$$t = \frac{\sqrt{42875}}{18} - \frac{3}{2} \quad \checkmark$$

$$= \underline{10 \text{ sec (to nearest s)}}$$

4) A) i.) R.T.P. $\frac{d}{dx} (\cos^3 x \sin x) = 4 \cos^4 x - 3 \cos^2 x$.

$$\text{LHS } \frac{d}{dx} (\cos^3 x \sin x) = \frac{d}{dx} [\sin x (1 - \sin^2 x) \cos x]$$

$$= \frac{d}{dx} \left(\frac{\sin 2x}{2} (1 - \sin^2 x) \right)$$

$$= \frac{1}{2} \frac{d}{dx} \sin 2x (\cos^2 x)$$

$$= \frac{1}{2} (\cos^2 x \cdot 2 \cos 2x + \sin 2x \cdot 2 \cos x \cdot -\sin x) = \frac{1}{2} (2 \cos^2 x (2 \cos^2 x - 1) - 2 \sin x \cos x \times 2 \sin x \cos x)$$

$$= \frac{1}{2} (4\cos^4 x - 2\cos^2 x - 4\sin^2 x \cos^2 x)$$

$$= \frac{1}{2} (4\cos^4 x - 2\cos^2 x - 4\cos^2 x (1 - \cos^2 x))$$

$$= \frac{1}{2} (4\cos^4 x - 2\cos^2 x - 4\cos^2 x + 4\cos^4 x) \quad \checkmark$$

$$= \frac{1}{2} (8\cos^4 x - 6\cos^2 x) = 4\cos^4 x - 3\cos^2 x = \underline{\text{RHS}}$$

ii.) $\int_0^{\frac{\pi}{4}} \cos^4 x \, dx = \frac{1}{4} + \frac{3\pi}{32}$

show that:

we know that $\int 4\cos^4 x - 3\cos^2 x \, dx = \cos^3 x \sin x + c$

$$\int 4\cos^4 x \, dx = \cos^3 x \sin x + \int 3\cos^2 x \, dx + c$$

$$\therefore \int \cos^4 x \, dx = \frac{\cos^3 x \sin x}{4} + \frac{1}{4} \int 3\cos^2 x \, dx + c \quad \checkmark$$

$$\int_0^{\frac{\pi}{4}} \cos^4 x \, dx = \left(\frac{\cos^3 x \sin x}{4} \right)_0^{\frac{\pi}{4}} + \frac{3}{8} \int_0^{\frac{\pi}{4}} (1 + \cos 2x) \, dx$$

$$= \frac{(\cos \frac{\pi}{4})^3 \sin \frac{\pi}{4}}{4} + \frac{3}{8} \left(x + \frac{1}{2} \sin 2x \right)_0^{\frac{\pi}{4}}$$

$$= \frac{(\frac{1}{\sqrt{2}})^4}{4} + \frac{3}{8} \left(\frac{\pi}{4} + \frac{1}{2} \sin \frac{\pi}{2} \right)$$

$$= \frac{1}{16} + \frac{3}{8} \left(\frac{\pi}{4} + \frac{1}{2} \right)$$

$$= \frac{1}{16} + \frac{3\pi}{32} + \frac{3}{16} = \frac{1}{4} + \frac{3\pi}{32} \quad \checkmark$$

$$\therefore \int_0^{\frac{\pi}{4}} \cos^4 x \, dx = \frac{1}{4} + \frac{3\pi}{32}$$

-7- (4)

$$\text{R.T.P. } 1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + 4 \times 2^3 + \dots + n \times 2^{n-1} = 1 + (n-1) 2^n$$

b) Step 1

let $n=1$

$$\underline{\text{LHS}} \quad n \times 2^{n-1} = 1 \times 2^0 = 1$$

$$\underline{\text{RHS}} \quad 1 + (n-1) 2^n = 1 + 0 = 1 = \text{LHS.} \quad \therefore \text{true for } n=1.$$

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Step 2

Assume true for $n=k$.

$$1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k \times 2^{k-1} = 1 + (k-1) 2^k$$

R.T.P. also true for $n=k+1$

$$1 \times 2^0 + 2 \times 2^1 + \dots + k \times 2^{k-1} + (k+1) 2^k = 1 + k \cdot 2^{k+1}$$

$$\underline{\text{LHS}} \quad 1 \times 2^0 + 2 \times 2^1 + \dots + k \times 2^{k-1} + (k+1) 2^k = 1 + (k-1) 2^k + (k+1) 2^k$$

(from assumption)

$$= 1 + 2^k (k-1 + k+1)$$

$$= 1 + 2^k (2k)$$

$$= 1 + 2^k \times 2 \times k = 1 + 2^{k+1} \cdot k = \underline{\text{RHS}}$$

Step 3

If true for $n=k$ and $n=k+1$ and also true for $n=1$, then it is true for $n=1+1=2$, $n=2+1=3$ and so on. $\therefore$ by P.O.M.I., it is true for all integers n .

c) $\cos 2x = \frac{1}{2}$

$$2x = \cos^{-1} \left(\frac{1}{2} \right)$$

$$2x = 2n\pi \pm \left(\frac{\pi}{3} \right)$$

$$x = n\pi \pm \frac{\pi}{6}$$

5)

a.) $y = x^2 - 1$

$$\frac{dy}{dx} = 2x \quad \text{--- } m_1$$

$$y = x^2 - x$$

$$\frac{dy}{dx} = 2x - 1 \quad \text{--- } m_2$$

$$\tan \theta = \left| \frac{2x - (2x - 1)}{1 + 2x(2x - 1)} \right|$$

$$= \left| \frac{2x - 2x + 1}{1 + 4x^2 - 2x} \right|$$

$$\tan \theta = \left| \frac{1}{4x^2 - 2x + 1} \right|$$

$$\tan \theta = \left| \frac{1}{4 - 2 + 1} \right| \sqrt{2} = \left| \frac{1}{3} \right|$$

$$\tan \theta = \frac{1}{3}$$

$$\theta = 18^\circ 26'$$

To find where the two curves intersect, solve $y = x^2 - 1$ and $y = x^2 - x$ simult.

$$x^2 - 1 = x^2 - x$$

$$x^2 - 1 - x^2 + x = 0$$

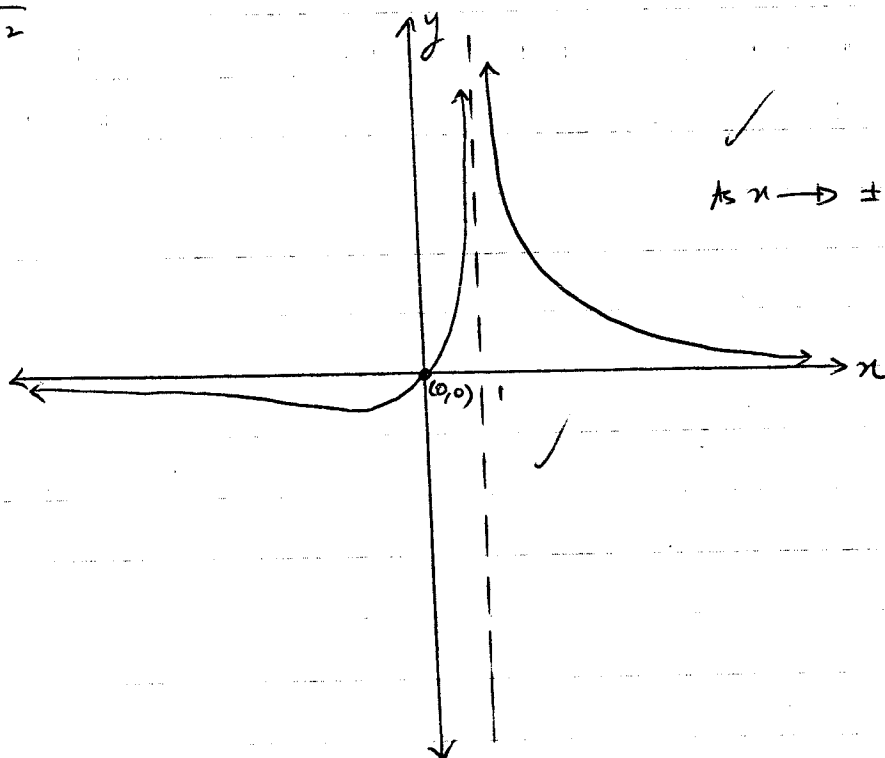
$$x - 1 = 0$$

$$x = 1$$

$\therefore$ they intersect at $x = 1$

(θ is acute. $\therefore \tan \theta > 0$)

b) $y = \frac{x}{(x-1)^2}$



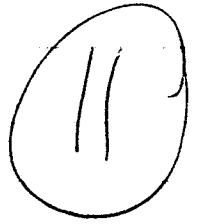
As $x \rightarrow \pm \infty$, $y \rightarrow 0$

(5)

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$$c) \quad i.) \quad P(\text{Joyce wins first draw}) = P(\text{even})$$

$$= \frac{3}{6} = \frac{1}{2} \quad \checkmark$$



$$ii.) \quad P(\text{J wins 2nd draw}) = P(\text{J loses 1st}) \times P(\text{A loses 1st}) \times P(\text{J wins 2nd})$$

$$= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8} \quad \checkmark$$

$$iii.) \quad P(\text{J wins 3rd draw}) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$$

$$= \frac{1}{32} \quad \checkmark$$

$$iv.) \quad P(\text{J wins}) = P(\text{A wins})$$

$$\therefore P(\text{J wins in long run}) = \frac{1}{2} + \frac{1}{8} + \frac{1}{32} + \dots \Rightarrow S_{\infty} = \frac{a}{1-r}$$

$$= \frac{\frac{1}{2}}{1 - \frac{1}{4}} = \frac{2}{3}$$

$$D) \quad \frac{dA}{dt} = 10 \text{ cm}^2/\text{s}. \quad \text{Find } \frac{dV}{dt} \text{ when } r = 5 \text{ cm}.$$

$$A = 4\pi r^2$$

$$\frac{dA}{dr} = 8\pi r.$$

$$\frac{dr}{dt} = \frac{dr}{dA} \times \frac{dA}{dt} = \frac{1}{8\pi r} \times 10 \quad \checkmark$$

$$= \frac{5}{4\pi r}$$

$$\therefore \frac{dr}{dt} = \frac{5}{4\pi r}.$$

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dr} = 4\pi r^2 \quad \checkmark$$

$$\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt} = \frac{4\pi r^2 \times 5}{4\pi r} = 5r$$

$$\frac{dV}{dt} = 5r \quad (r = 5)$$

$$\therefore \frac{dV}{dt} = 25 \text{ cm}^3/\text{s} \quad \checkmark$$

6) A.) i.) $x^2 = 16y$; $y = \frac{x^2}{16}$; $y' = \frac{x}{8}$.

At $P(8t, 4t^2)$, grad. of $tyt = \frac{8t}{8} = t$. ✓

Eqn of tyt is $(y - 4t^2) = t(x - 8t)$

$y = tx - 8t^2 + 4t^2$ ✓

$y = tx - 4t^2$

ii.) Focal length; $4A = 16$

$A = 4$

∴ since vertex of parabola is $(0, 0)$,

Focus = $(0, 4)$

∴ $S = (0, 4)$

grad of $SP = \frac{4t^2 - 4}{8t} = \frac{t^2 - 1}{2t}$ ✓

Since $SP \perp l$, grad. of $l = \frac{-2t}{t^2 - 1}$ ($m_1 m_2 = -1$)

Eqn of l is =

$(y - 4) = \frac{-2t}{t^2 - 1} (x - 0)$ ✓

$(t^2 - 1)(y - 4) = -2tx$

$y(t^2 - 1) - 4(t^2 - 1) + 2tx = 0$ ✓

∴ $y(t^2 - 1) + 2tx = 4(t^2 - 1)$

iii.) Point of intersection of line l and tyt at P is given by

$tx - 4t^2 = \frac{4(t^2 - 1) - 2tx}{t^2 - 1}$

$(t^2 - 1)(tx - 4t^2) = 4(t^2 - 1) - 2tx$

$(t^2 - 1)(tx - 4t^2 - 4) = -2tx$

$t^3x - 4t^4 - 4t^2x + 4 + 2tx = 0$

$x(t^3 - t + 2t) - 4t^4 + 4 = 0$

$x = \frac{4t^4 - 4}{t^3 - t + 2t}$

$y = t \left(\frac{4t^4 - 4}{t^3 - t + 2t} \right) - 4t^2$

(6)

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iii) Eq of tgr at P is $y = tx - 4t^2$

$$y + 4t^2 - tx = 0$$

$$4t^2 - tx + y = 0$$

$$t = \frac{x \pm \sqrt{x^2 - 4(4)(y)}}{8} = \frac{x \pm \sqrt{x^2 - 16y}}{8}$$

Since P lies on parabola, $x^2 = 16y$, $x^2 - 16y = 0$.

$$t = \frac{x \pm \sqrt{16y - 16y}}{8}$$

$$t = \frac{x}{8} \quad (\text{sub into } l) \quad \checkmark$$

$$\text{Eq of } l \text{ is } = (t^2 - 1)y + 2tx = 4(t^2 - 1)$$

$$(t^2 - 1)(y - 4) + 2tx = 0$$

$$\left(\frac{x^2}{64} - 1\right)(y - 4) + 2x\left(\frac{x}{8}\right) = 0 \quad \checkmark$$

$$\left(\frac{x^2 - 64}{64}\right)(y - 4) + \frac{x^2}{4} = 0$$

$$y - 4 = -\frac{x^2}{4} \quad ; \quad y - 4 = -\frac{x^2}{4} \times \frac{64}{x^2 - 64}$$

$$y - 4 = -\frac{16x^2}{x^2 - 64}$$

$$y = 4 - \frac{16x^2}{x^2 - 64} \quad \checkmark$$

$$y = \frac{4x^2 - 256 - 16x^2}{x^2 - 64}$$

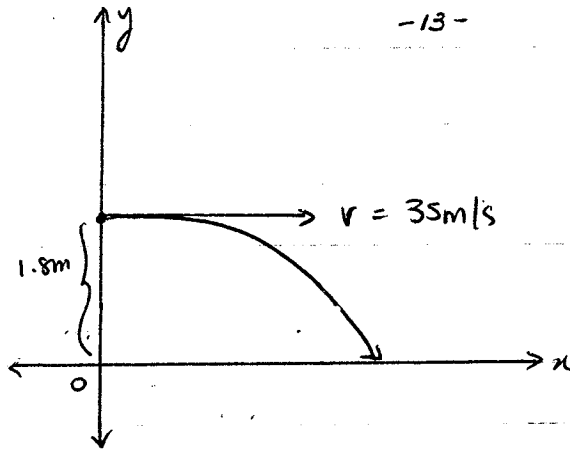
$$\underline{\underline{y = \frac{-12x^2 - 256}{x^2 - 64} \quad \checkmark}}$$

(7)

-13-

7)

A)



Given:

$$y = -5t^2 + 1.8$$

$$\dot{y} = -10t$$

$$x = 35t$$

$$\dot{x} = 35$$

i.) Find t when $y = 0$.

$$1.8 - 5t^2 = 0 \quad \checkmark$$

$$5t^2 = 1.8$$

$$t^2 = 0.36$$

$$t = 0.6 \quad (t > 0) \quad \checkmark$$

$\therefore$ it takes 0.6s for ball to hit ground.

ii.) When $t = 0.6$, $x = 35(0.6)$

$$= 21\text{m} \quad \checkmark$$

$\therefore$ the ball travels 21m before bouncing.

iii.) When $x = 14$, $35t = 14$

$$t = \frac{14}{35} = \frac{2}{5} \text{ s} \quad \checkmark$$

$$\text{When } t = \frac{2}{5} \text{ s, } y = -5\left(\frac{4}{25}\right) + 1.8$$

$$= 1\text{m} > 0.95\text{m.} \quad \checkmark$$

$\therefore$ the ball WILL clear the net, by 0.05m

(12)

B) i.) $DC = DP = DQ$ (pts from exterior pt, equal)

$\therefore \triangle PDC$ and $\triangle CDQ$ are isos $\triangle$.

Let $\angle CPD = \alpha$. $\angle CPD = \angle PCD = \alpha$ (base $\angle$ of isos $\triangle$ equal)

$\angle CDQ = 2\alpha$ (exterior $\angle$ equals sum of two interior opp $\angle$)
 $\angle DCQ = \frac{180^\circ - 2\alpha}{2}$ ($\angle$ sum $\triangle = 180^\circ$, base $\angle$ of isos $\triangle$ equal)

$$= 90^\circ - \alpha$$

$$\angle PCQ + \angle PCD = 90^\circ - \alpha + \alpha = 90^\circ$$

$$\therefore \angle PCQ = 90^\circ$$

ii.) $\angle PCQ = 90^\circ$, (already proven)

$\angle QCR = 90^\circ$ ($\angle$ in semicircle with radius $RQ = 90^\circ$)

$$\angle PCR = \angle PCQ + \angle QCR$$

$$= 90^\circ + 90^\circ = 180^\circ$$

$\therefore PCR$ is a straight line.

$\therefore P, C$, and R are collinear

iii.) $\angle CPD = \alpha$.

$\therefore \angle APC = 90^\circ - \alpha$ ($\angle APD = 90^\circ$, $\angle$ made by radius

$\angle TCP = 90^\circ$ ($\angle$ in semicircle $= 90^\circ$) and $\angle T = 90^\circ$)

$$\therefore \angle PTC = 180^\circ - (90^\circ - \alpha) - 90^\circ \quad (\angle \text{sum } \triangle = 180^\circ)$$

$$= \alpha$$

$$\angle PTT = \angle PSC = \alpha$$

$$\text{Since } \angle APC = 90^\circ - \alpha, \quad \angle TSC = 180^\circ - (90^\circ - \alpha) \quad (\text{opp } \angle \text{ of cyclic quad } PTSC \text{ are supp})$$

$$= 90^\circ + \alpha$$

$$\therefore \angle CSR = 180^\circ - (90^\circ + \alpha) \quad (\text{straight line} = 180^\circ)$$

$$= 90^\circ - \alpha$$

$$\angle PSR = \angle PSC + \angle CSR$$

$$= \alpha + 90^\circ - \alpha = 90^\circ$$

$$\therefore \angle PSR = 90^\circ$$

$$\angle PSR + \angle PQR = 90^\circ + 90^\circ = 180^\circ$$

$\therefore PQRS$ is a cyclic quad (opp $\angle$ of cyclic

$\therefore P, Q, R$ and S are concyclic points, quad are supp)